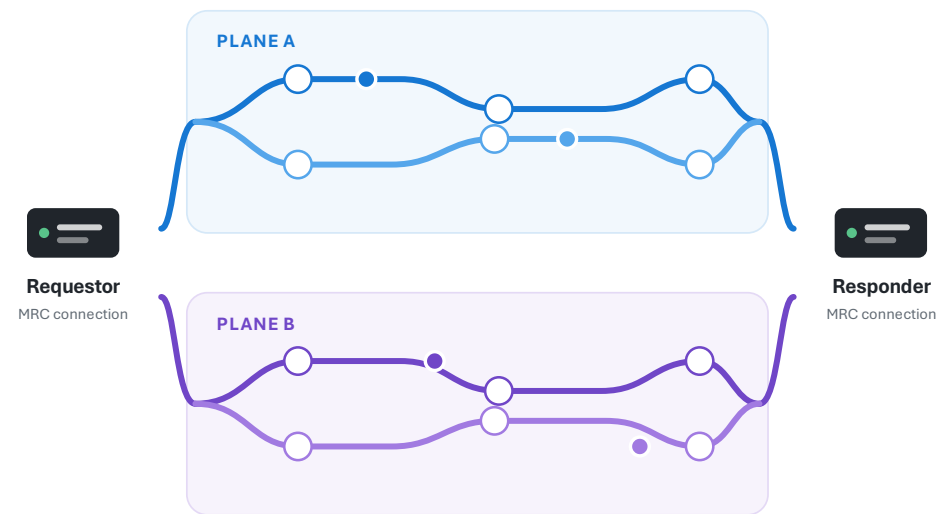


The Multipath Reliable Connection (MRC) Transport

Reliable multipath and multiplane transport over best-effort Ethernet



Rip Sohan, Eric Spada, Eric Davis, Mark Handley, Idan Burstein, Tony Hurson, Jithin Jose, Vivek Kashyap, Rong Pan, Sayantan Sur, Sreevatsa Anantharamu, Aviv Barnea, Adrian Caulfield, Elazar Cohen, Elliot Edmunds, Yamin Friedman, Mahdieh Ghazi, Murali Garimella, Torsten Hoeffler, Vipin Jain, Abdul Kabbani, Noam Katz, Yanfang Le, Charlie Mbariky, Guglielmo Morandin, Masoud Moshref, Shane O'Neil, Michael Papamichael, Jonas Pfefferle, Siva Santosh Pyla, Costin Raiciu, David Riddoch, Karen Schramm, Yuval Shpigelman, Shahaf Shuler, Shy Shyman, Raghava Sivaramu, Amin Tootoonchian, Yang Wang

Hot Interconnects, 2026



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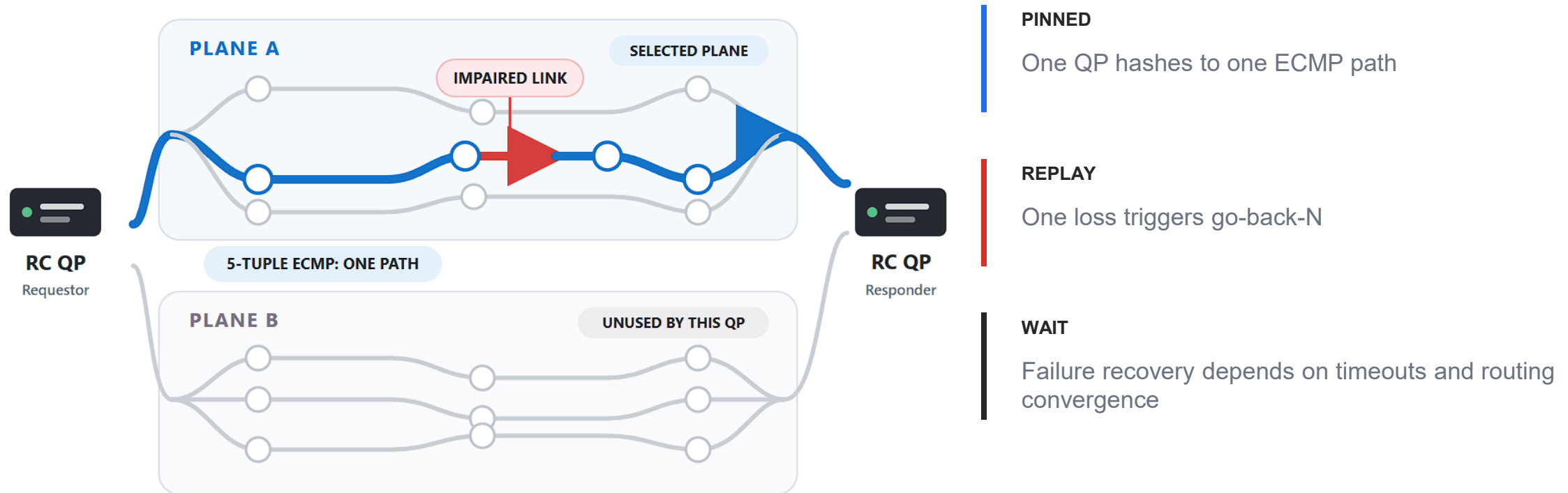
Microsoft



NVIDIA

OpenAI

Limitations of RC at AI Scale



MRC Origins and Design Objectives

EARLY 2024



Proven foundations

Ultra Ethernet Consortium work
and earlier vendor designs for
multipath and packet spraying

SHARED AMBITION

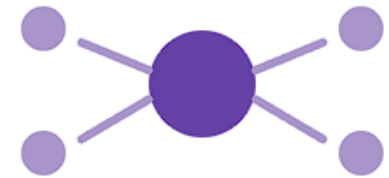
10^6 nodes

DESIGN DIRECTION

Diversity + resilience

OpenAI and Microsoft sought
multipath and multiplane operation
resilient to link and fabric failures

OPEN COLLABORATION



One open specification

AMD · Broadcom · Intel
Microsoft · NVIDIA · OpenAI

OCP MRC



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intel



Microsoft



nVIDIA

OpenAI

Application Semantics and Transport Mechanisms

APPLICATION-VISIBLE CONTRACT



Reliable delivery

Connection-level reliability



Ordered completions

Disorder remains below the API



Familiar model

QP/CQ-style software abstractions

PRESERVED

MRC TRANSPORT MACHINERY



Spray

Per packet



Bound

Inflight state



Recover

Selectively



Control

Sender + paths



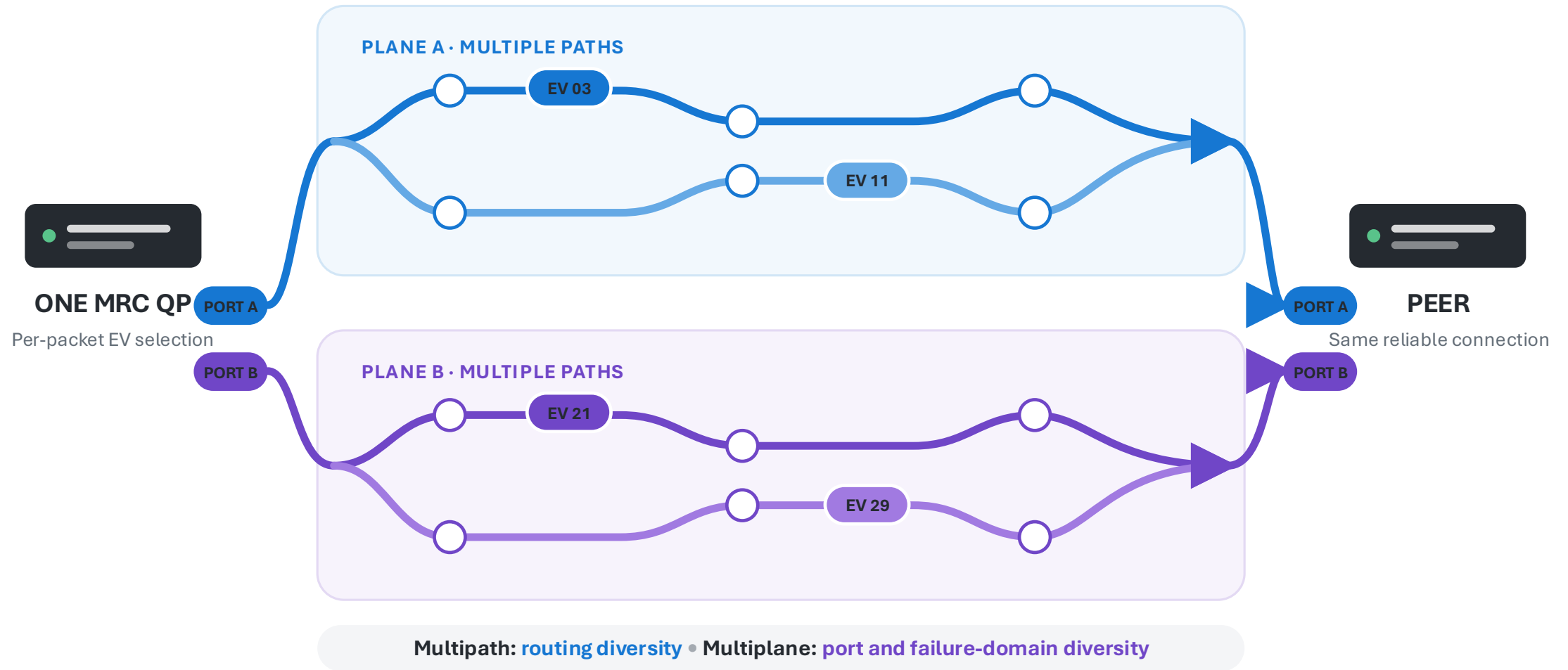
Avoid

Failed paths

Mandatory primitives ensure correctness • Optional primitives add efficiency and resilience



Multipath and Multiplane



Source Routing With SRv6

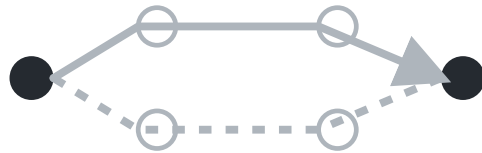
ONE MRC ENTROPY VALUE ABSTRACTION

The transport can use implicit hashing or explicit source routes

CONVENTIONAL

ECMP entropy

Endpoint selects a hash input

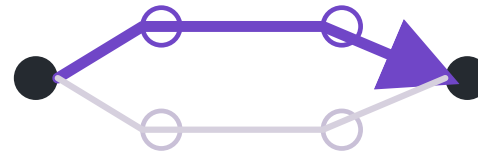


PATH CHOSEN BY THE NETWORK

EXPLICIT

Structured EV

Hop-by-hop route encoded in packet fields



ROUTE IDENTIFIED BY EV

EXPLICIT

SRv6 uSID

Path encoded as IPv6 micro-segments



ROUTE IDENTIFIED BY EV

WHEN A ROUTE DEGRADES



Identify

Feedback maps to route



Exclude

Denylist unhealthy EV



Reselect

Use another known route

NO NEED TO WAIT FOR
ROUTING CONVERGENCE



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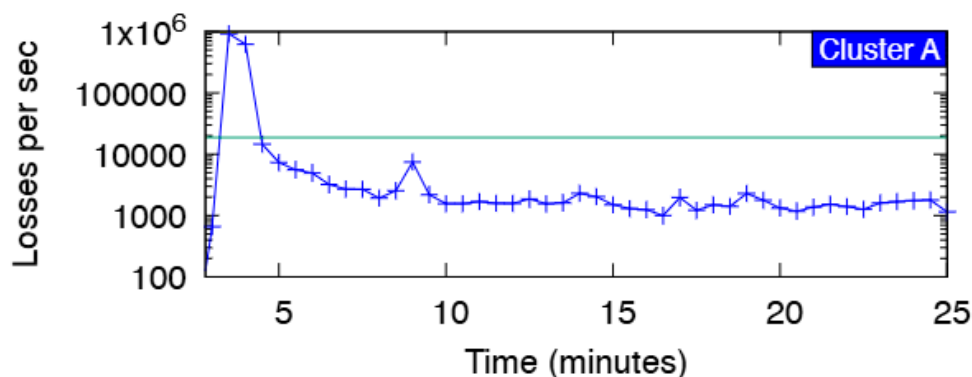
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OpenAI

Production Results: Automatic Path Avoidance at 75K GPUs

PRODUCTION PRETRAINING STARTUP

Empty initial denylist • Explicit SRv6 routes • Losses aggregated across the job



Source: Resilient AI Supercomputer Networking using MRC and SRv6, Fig. 4

PRODUCTION SCALE

75K GPUs

pretraining job

INITIAL DISCOVERY

<5 losses / QP

first minute

WITHIN MINUTES

<1 loss/s/NIC

≈ one packet in 25 million at 800 Gb/s

Observe failed delivery



Retransmit selectively



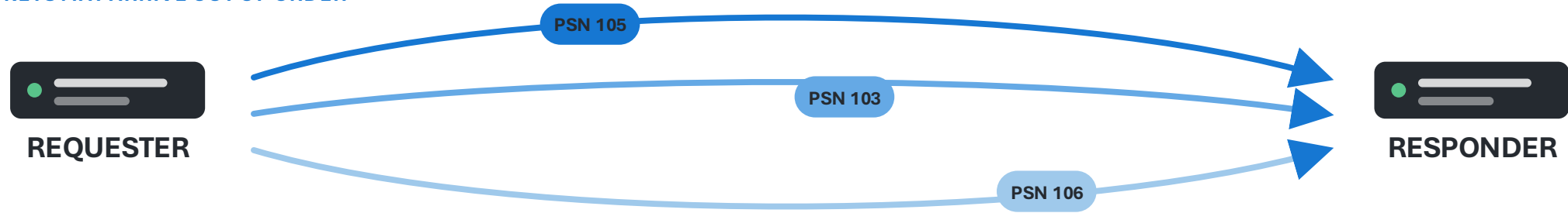
Remove failed EV



Continue on backup route

Bounded Reordering

PACKETS MAY ARRIVE OUT OF ORDER



RESPONDER-ADVERTISED MAXIMUM PSN RANGE

MPR = 8 packets



Bounded requester state

Retransmission tracking ≤ MPR

Bounded responder state

Out-of-order tracking ∝ MPR

Ordered completion preserved

Disorder remains below the API

Optional Dynamic MPR resizes the window as demand and available NIC resources change



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Selective Loss Detection and Recovery

CONVENTIONAL RC

Go-back-N replays the tail

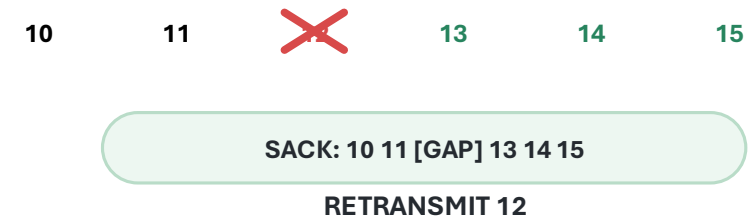
One missing packet invalidates later progress



MRC

Selective recovery preserves progress

SACK identifies received packets and the actual gap



DETECT EARLY · RECOVER PRECISELY · GUARANTEE PROGRESS



SACK + NACK

Explicit delivery state

Mandatory reliability feedback



Packet trim

Header survives the drop

Optional network acceleration



Probe

Elicit current SACK state

Optional proactive query



Timers + priority

Fallback and fast repair

Oldest gap retransmitted first



Congestion Control Framework and NSCC

STANDARDIZED BY MRC

SACK-carried feedback

One feedback stream closes multiple loops

- CE** **ECN state**
Forward-path congestion marking
- ✓** **Delivery accounting**
SACK clock and cumulative bytes received
- 🕒** **Network RTT**
RTT minus optional responder service time
- M** **Responder backpressure**
Host and memory-subsystem congestion
- EV** **Reflected Entropy Value**
Correlates feedback with a specific path

PER-CONNECTION POLICY

CC PROFILE

Selects supported policy

Configured per connection
by the privileged controller

TRANSPORT DEFINES THE SIGNALS

Implementation defines how to react

IMPLEMENTATION BEHAVIOR

USED IN THIS IMPLEMENTATION

NSCC

Network Signaled Congestion Control

Sender-based · window-driven · SACK-clocked

Other algorithms

May consume all or a subset of MRC signals

CONTROL ACTIONS

Byte window
NSCC

EV selection
Load balancer

SEND → MEASURE → ACK → ADJUST



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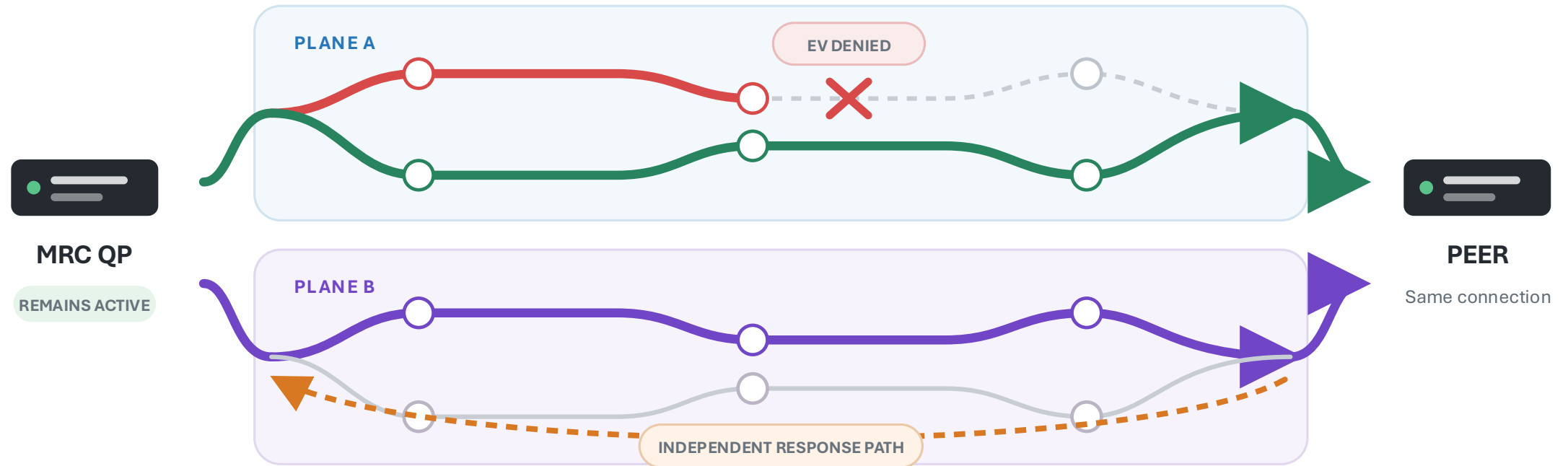
Microsoft



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Fast Path and Plane Failure Avoidance



DATAPATH-TIMESCALE RESILIENCE



EV state

Exclude unhealthy routes

Path-selection state



EV probe

Test one explicit path

Path-scoped liveness



Port Status

Announce plane reachability

Asynchronous update



GID scoped

Update many QPs at once

Stateless endpoint operation



Application and Controller APIs

APPLICATION API

`mrc.h`

Familiar verbs model

Transport control and data-plane operations

- ✓ **Reuse libibverbs**
Discovery and memory registration
- ✓ **Familiar resources**
`mrc_qp` and `mrc_cq` lifecycle
- ✓ **Discover capabilities**
Device and per-port feature queries
- ✓ **Configure and transmit**
Write and Write-with-Immediate

`query_device · query_port · modify_qp`

OPERATE



DATA PLANE

Reliable connection

BINDS TO

EV PROFILE

CC PROFILE

Connection setup remains out of band

Best-effort Ethernet

RoCEv2 / UDP / IP encapsulation

POLICY



CONTROLLER API

`mrc_ctl.h`

Shared transport policy

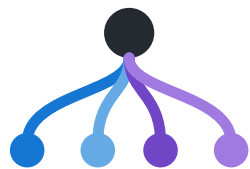
Privileged process with `CAP_NET_ADMIN`

- ✓ **Query capabilities**
Mandatory device and port discovery
- ✓ **Create EV profiles**
Shared path universe and selection policy
- ✓ **Create CC profiles**
Shared congestion-control policy
- **Optional path controls**
EV-state events and EV probes

Policy and telemetry shared across many QPs

MRC: Open, Production-Deployed, and Available

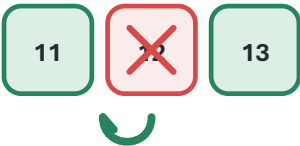
USE THE DIVERSITY



Multipath + multiplane

One QP uses routing, port, and failure-domain diversity

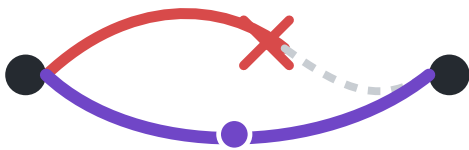
PRESERVE PROGRESS



Reliable on best effort

Bound disorder and recover only what was actually lost

KEEP MOVING



Resilient by construction

Avoid failed routes without waiting for convergence



Open OCP specification

A cross-industry transport specification



Already in production

Production-grade—not a paper design



Standard encapsulation

RoCEv2 / UDP / IP over Ethernet

opencompute.org/documents/ocp-mrc-1-0-pdf